Houssam Zenati

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Research Interests

I develop statistical theory and algorithms for reliable inference and decision-making with learned models. My work connects nuisance-robust inference, policy learning and adaptive experimentation: controlling the estimation errors that affect a target or decision, and collecting the observations needed to assess it. I am particularly interested in applying these ideas to biology and biomedicine.

Professional Experience

Gatsby Computational Neuroscience Unit, University College London, United Kingdom
Research Fellow 2025–present
Developing methods for semiparametric efficient distributional causal inference, adaptive data collection, and sequential decision-making.

INRIA MIND / PreMeDical, France
Postdoctoral Researcher 2023–2024
Research on causal inference and mediation analysis, including double machine learning estimators and open-source software development.

Criteo AI Lab, France
PhD Researcher 2020–2023
Research Engineer 2019–2020
Worked on counterfactual policy learning and online/bandit problems, including optimization methods, contextual bandits, and kernel methods.

Institute for Infocomm Research (A*STAR), Singapore 2017–2018
Research Engineer
Research Intern
Worked on deep representation learning, generative adversarial networks, anomaly detection, and semi-supervised learning, with applications to medical imaging.

Education

Université Grenoble Alpes / INRIA Thoth, France 2023
Ph.D. in applied mathematics and machine learning.
Thesis: *Efficient Methods in Counterfactual Policy Learning and Sequential Decision Making.*

ENS Paris-Saclay / MVA, France 2019
M.Sc. in mathematics, vision, and learning.

CentraleSupélec, France 2015–2019
M.Eng. in applied mathematics and computer science.

Selected Publications

$*$ marks authors who contributed equally to the paper.

Z. Shen*, **H. Zenati***, N. Kallus, A. Gretton, K. Khamaru, A. Bibaut (2026). *Efficient Inference after Directionally Stable Adaptive Experiments.* Preprint.
arXiv:2602.21478

H. Zenati, B. Bozkurt, A. Gretton (2026). *Kernel Treatment Effects with Adaptively Collected Data.* AISTATS.
AISTATS 2026 proceedings

H. Zenati, A. Gretton (2026). *Semiparametric Efficient Tests for Interpretable Distributional Treatment Effects.* NeurIPS.
arXiv:2605.08034

S. Girard, A. Bibaut, J.-J. Vie, A. Gretton, N. Kallus, **H. Zenati** (2026). *Fast Best-in-Class Regret for Contextual Bandits.* UAI.
UAI 2026 proceedings

H. Zenati, E. Diemert, M. Martin, J. Mairal, P. Gaillard (2023). *Sequential Counterfactual Risk Minimization.* ICML.
arXiv:2302.12120

Other Publications

∗ marks authors who contributed equally to the paper. † marks alphabetical author order.

R. Leman, B. Bozkurt, S. Tanji, A. Gretton, **H. Zenati** (2026). *Doubly Robust Deep Kernel Tests for Structured Outcomes*. Preprint.

F. El Khoury*, **H. Zenati***, N. Kallus, M. Arbel, A. Bibaut (2026). *Parametric Rates for Nonparametric Conditional-Moment Bilevel Optimization*. Preprint.

B. Bozkurt, A. Galashov, D. Meunier, Z. Shen, A. Gretton, **H. Zenati** (2026). *Doubly Robust Proxy Causal Learning with Neural Mean Embeddings*. Preprint.

Z. Shen, N. Kallus, D. Meunier, **H. Zenati**, A. Gretton, A. Bibaut (2026). *Nonparametric Instrumental Variable Analysis Without Structural Equations: Debiased Inference on Functionals of Inverse Problems with No Solutions*. Preprint.

A. Bibaut, **H. Zenati**, T. Rahier, N. Kallus (2026). *Functional Natural Policy Gradients*. Preprint.

H. Zenati, B. Bozkurt, A. Gretton (2025). *Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings*. NeurIPS.

B. Bozkurt, **H. Zenati**, D. Meunier, L. Xu, A. Gretton (2025). *Density Ratio-Free Doubly Robust Proxy Causal Learning*. NeurIPS.

H. Zenati, J. Abecassis, J. Josse, B. Thirion (2025). *Double Debiased Machine Learning for Mediation Analysis with Continuous Treatments*. AISTATS.

J. Abecassis, **H. Zenati**, S. Boumaiza, J. Josse, B. Thirion (2025). *Causal Mediation Analysis with One or Multiple Mediators: A Comparative Study*. Psychological Methods.

H. Zenati, A. Bietti, M. Martin, E. Diemert, P. Gaillard, J. Mairal (2025). *Counterfactual Learning of Stochastic Policies with Continuous Actions*. Transactions on Machine Learning Research.

J. Zhou, P. Gaillard, T. Rahier, **H. Zenati**, J. Arbel (2024). *Towards Efficient and Optimal Covariance-Adaptive Algorithms for Combinatorial Semi-Bandits*. NeurIPS.

M. Martin, P. Mertikopoulos, T. Rahier, **H. Zenati** (2022). *Nested Bandits*†. ICML.

H. Zenati, A. Bietti, E. Diemert, J. Mairal, M. Martin, P. Gaillard (2022). *Efficient Kernelized UCB for Contextual Bandits*. AISTATS.

H. Zenati, A. Bietti, M. Martin, E. Diemert, J. Mairal (2020). *Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions*. ICLR Workshop on Causal Learning for Decision Making.

P. Mertikopoulos, B. Lecouat, **H. Zenati**, C.-S. Foo, V. Chandrasekhar, G. Piliouras (2019). *Optimistic Mirror Descent in Saddle-Point Problems: Going the Extra (Gradient) Mile*. ICLR.

K. Ouardini, H. Yang, B. Unnikrishnan, M. Romain, C. Garcin, **H. Zenati**, J. P. Campbell, M. F. Chiang, J. Kalpathy-Cramer, V. Chandrasekhar, et al. (2019). *Towards Practical Unsupervised Anomaly Detection on Retinal Images*. MICCAI Workshop on Domain Adaptation and Representation Transfer.

H. Zenati, M. Romain, C.-S. Foo, B. Lecouat, V. Chandrasekhar (2018). *Adversarially Learned Anomaly Detection*. ICDM.

H. Zenati, C.-S. Foo, B. Lecouat, G. Manek, V. R. Chandrasekhar (2018). *Efficient GAN-Based Anomaly Detection*. ICLR Workshop.

B. Lecouat, C.-S. Foo, **H. Zenati**, V. Chandrasekhar (2018). *Semi-Supervised Learning with GANs: Revisiting Manifold Regularization*. ICLR Workshop.

B. Lecouat, C.-S. Foo, **H. Zenati**, V. Chandrasekhar (2018). *Manifold Regularization with GANs for Semi-Supervised Learning*. arXiv preprint.

B. Lecouat, K. Chang, C.-S. Foo, B. Unnikrishnan, J. M. Brown, **H. Zenati**, A. Beers, V. Chandrasekhar, J. Kalpathy-Cramer, P. Krishnaswamy (2018). *Semi-Supervised Deep Learning for Abnormality Classification in Retinal Images*. NeurIPS Workshop on Machine Learning for Health.

Teaching

Data Visualisation, Université Paris Dauphine – PSL 2021–2022
Instructor for Master's students in the IASD program. Covered visualisation as a statistical and algorithmic tool for exploring, summarizing, and communicating complex data, with topics including graphical principles and dimensionality-reduction methods such as PCA, multidimensional scaling, and t-SNE.

Advanced Machine Learning, CentraleSupélec 2024–2025
Teaching assistant alongside Emilie Chouzenoux. Helped students connect the formal ideas in modern machine learning, statistics, and optimization to practical algorithms and empirical behavior.

Causal Inference, New York University Paris 2024–2025
Co-instructor with Judith Abecassis. Taught four sessions, including lectures and recitations, on randomized and observational causal inference, including identification assumptions, weighting, matching, instrumental variables, and regression discontinuity.

Teaching philosophy: conceptual clarity, mathematical rigor, and active engagement.

Student Supervision

Research co-supervision and mentoring at Master's and doctoral levels.

Julien Zhou, Master's student, with Pierre Gaillard 2023
Co-supervised research on covariance-adaptive combinatorial semi-bandits, leading to a NeurIPS 2024 publication.

Samuel Girard, PhD student, with Aurélien Bibaut 2025–2026
Mentored research on contextual bandits, leading to a UAI 2026 publication.

Robin Leman, Master's student, with Arthur Gretton 2026
Research supervision and mentoring.

Bariscan Bozkurt, PhD student, with Arthur Gretton 2026
Research mentoring.

Sofiane Tanji, PhD student, UCLouvain 2026
Co-supervising with Yassine Laguel.

Service

Workshop organisation
Lead Organiser, Advances in Adaptive Experimentation (AAE) Workshop, Gatsby Computational Neuroscience Unit, UCL, 18–19 June 2026. Two-day programme with seven invited speakers, a poster session and an open-problems session. Organising committee: Houssam Zenati, Bariscan Bozkurt, I-Chun Lin and Arthur Gretton.

Seminar organisation
Organiser of Gatsby's external machine learning seminars, UCL, September 2025–August 2026.

Reading groups
Co-organised, with Dimitri Meunier, a reading group on van der Vaart's *Asymptotic Statistics*, Gatsby, UCL, 2026.
Co-organised, with Fateme Ghayem, a reading group on Wainwright's *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Inria Paris, 2023–2024.

Peer review
NeurIPS, ICML, ICLR, AISTATS and TMLR.

Funding and Awards

Netflix workshop grant 2026
Secured US$5,000 from Netflix to support the Advances in Adaptive Experimentation (AAE) Workshop at Gatsby, UCL, 18–19 June 2026.

IEEE ICDM Student Travel Award 2018
Support for conference participation.

Singapore International Pre-Graduate Award (SIPGA) 2017
Support for a research attachment at A*STAR, Singapore.

Talks

Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings, INRIA Thoth, France — Oct. 2025

Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings, Symposium on Mathematical Foundations of Trustworthy Learning, Switzerland — Oct. 2025

Sequential Counterfactual Risk Minimization, ELLIS Unconference, France — Jul. 2023

Sequential Counterfactual Risk Minimization, INRIA Lille, France — Jun. 2023

Nested Bandits, ICML, United States — Jul. 2022

Efficient Kernelized Contextual Bandits, virtual talk — May 2022

Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions, virtual talk — Apr. 2020

Adversarially Learned Anomaly Detection, ICDM, Singapore — Nov. 2018

Efficient GAN-Based Anomaly Detection, DL2.0 Workshop, I2R, A*STAR, Singapore — Apr. 2018

Software

med_bench: Python package for causal mediation analysis with a unified implementation of common estimators and flexible nuisance models.

Open-source research code:

- interpretable-dr-me-test
- adaptive-KTE
- counterfactual-policy-mean-embedding
- double-debiased-machine-learning-mediation-continuous-treatments
- sequential-counterfactual-risk-minimization
- Efficient-Kernel-UCB
- Nested-Exponential-Weights
- optimization-continuous-action-crm
- Adversarially-Learned-Anomaly-Detection
- Efficient-GAN-Anomaly-Detection